GoGoX: Optimizing logistics services with real-time data prediction

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By leveraging the
infrastructure an
processing tools
Cloud, GoGoX is
continuously imp
logistics services
real-time data pre
while enhancing
infrastructure sta
minimal operatio

About GoGoX

Founded in 2013, GoGoX is a leading logistics technology company in Asia with its footprint across more than 360 cities and a network of more than 6.1 million registered drivers in six Asian regions, including Singapore, Vietnam, India, South Korea, Mainland China, and Hong Kong. Offering extensive logistics services through innovative technology, from customized logistics solutions to instant deliveries and value-added services, such as fuel card and insurance, GoGoX is now listed on the main board of the Hong Kong Stock Exchange.

Google Cloud results

- Reduces operational workloads and ensures high infrastructure reliability with GKE
- Enables easy data pipeline orchestration from various sources through Cloud Composer
- Supports serverless real-time data streaming processing with Pub/Sub and Dataflow
- Helps enhance delivery efficiency and drivers' earnings with predictive AI

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Industries: Transportation & Logistics

Location: Hong Kong

Composer

Google Cloud (https://cloud.google.com/)

BigQuery (https://cloud.google.com/bigquery/)

Cloud Composer (https://cloud.google.com/composer/)

Cloud SQL (https://cloud.google.com/sql/)

Dataflow (https://cloud.google.com/dataflow/)

Kubernetes Engine (https://cloud.google.com/kubernetes-engine/)

Google Maps Platform (https://mapsplatform.google.com/)

Looker Studio (https://cloud.google.com/looker-studio)

Cloud Pub/Sub (https://cloud.google.com/pubsub/)

Google Workspace (https://workspace.google.com/)

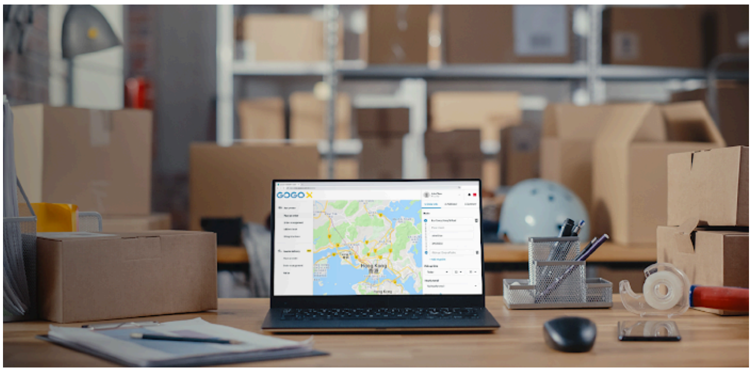
Sheets (https://workspace.google.com/products/sheets/)

models supported by BigQuery

The logistics industry around the world is changing rapidly with digitization (https://www.pwc.com/sg/en/publications/assets/future-of-the-logistics-industry.pdf). Leveraging the latest technologies, logistics service providers are striving to offer faster and more trustworthy deliveries to customers, as well as enable drivers to work more efficiently and smartly.

GoGoX (https://www.gogox.com/index/) is one of these innovation-oriented pioneers. Founded in 2013, the logistics technology company offers an extensive range of services, from customized logistics solutions to instant deliveries and value-added services, such as fuel card and insurance. Headquartered in Hong Kong, it has expanded its footprint across more than 360 cities in Singapore, Vietnam, India, South Korea, and Mainland China with a network of more than 6.1 million registered drivers.

"Our mission is to simplify logistics with technology," notes Fu-Ming Tsai, Head of Engineering at GoGoX. "By learning customer behavior and industry data, we provide tailor-made logistics solutions to our customers from individuals, SMEs to large corporations. That's why technology advancement is one of the key pillars for achieving our mission."



Ease of using GoGoX online

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> Besides, it offers powerful data processing tools that can help us realize our real-time data prediction project successfully."
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> *—Fu-Ming Tsai, Head of Engineering, GoGoX*

GoGoX first deployed its logistics service platform in a public cloud environment but in 2019 decided to migrate its service to Google Cloud (https://cloud.google.com/), because of its developer friendly interface and advanced containerization features. On top of that, to improve its services, GoGoX was planning to build predictive machine learning (ML) models for various aspects of its operations from driver behavior, driver earnings, to order completion and working time.

For more accurate and timely prediction, the GoGoX team needed to establish a centralized data pipeline that could process data from multiple sources in real time cost-effectively to train its ML models. Google Cloud also possesses a full set of data processing tools that meet the team's needs.

"We wanted to use the latest Kubernetes container technologies, but we didn't have the manpower to manage them ourselves. As the main contributor of Kubernetes, Google Cloud provides the best container solution," explains Tsai. "Besides, it offers powerful data processing tools that can help us realize our real-time data prediction project successfully."

Ensuring infrastructure stability while reducing operational workloads

GoGoX smoothly completed its migration to Google Cloud within two months in mid-2020. By using different Google Cloud projects to isolate its test and deployment environments, the GoGoX team was able to limit the blast radius of any problems that arose during the migration process to avoid impact on user experience. Throughout the process, Google Cloud engineers also provided timely responses to technical issues to facilitate the migration.

Now, GoGoX uses Kubernetes container clusters hosted in Google Kubernetes Engine (https://cloud.google.com/kubernetes-engine) (GKE) to run its logistics service platform, saves its platform data in Cloud SQL (https://cloud.google.com/sql), and provides route planning on its drivers' app with Google Maps Platform (https://mapsplatform.google.com/). Tsai says that the advanced container technology of GKE enables autoscaling and flexible resource allocation. It is also easy to roll back to the previous software version during deployment with GKE. All these features have together helped the team save substantial manual work and ensure high infrastructure stability.

"Containers are the central element of our IT infrastructure, and GKE enables us to manage our container clusters at ease and ensures the availability of underlying resources. Since the migration to Google Cloud, we haven't encountered any major incidents resulting from our infrastructure," he adds.

Building a centralized data pipeline for higher data processing efficiency and decision making

GoGoX has long been relying on data to make business decisions. It employs multiple systems and pipelines to retrieve data from various sources like mobile apps, internal databases, and third-party services into its data warehouse, which requires dedicated manpower to ensure data integrity. On a daily basis, teams in separate departments have their own way of working. Some would key in their data on spreadsheets hosted offline, while others would key them into Sheets (https://workspace.google.com/products/sheets/) hosted on Google Workspace (https://workspace.google.com/intl/en_sg/), making it challenging to incorporate data from multiple sources into dashboards and reports.

After the migration to Google Cloud, GoGoX started using Cloud Composer (https://cloud.google.com/composer) to host extract, transform and load (ETL) jobs from various data sources. As Cloud Composer supports one-click deployment, the GoGoX team can orchestrate its data workflows effortlessly. The end-to-end integration of Cloud Composer with other Google Cloud tools has also enabled the team to easily build a centralized ETL data pipeline with BigQuery (https://cloud.google.com/bigquery) as its data warehouse and Looker Studio (https://cloud.google.com/looker-studio) for statistics visualization.

"Our business data come from various sources, so it was labor- and time-consuming to process all the data for decision-making. With Google Cloud, it's a lot easier to incorporate such data and give prompt and actionable insight to different stakeholders," says Tsai.

Optimizing logistics services through real-time data prediction

GoGoX has a strong relationship with drivers and is always looking for ways to help them earn more. In its ongoing order assignment (OA) project, the company leverages real-time predictions for OA driver expected earnings and GoGoX business (GGB) order demand. The goal is to empower OA drivers to complete as many GGB orders as possible while ensuring they earn at least the guaranteed income.

This intricate balance between maximizing order fulfillment and maintaining financial security for drivers is achieved through advanced machine learning (ML) algorithms. GoGoX applies the capture data change (CDC) concept to its ETL pipeline to obtain data in real time for ML model training. To that end, it employs Pub/Sub (https://cloud.google.com/pubsub) as its message broker and Dataflow (https://cloud.google.com/dataflow) to stream data processing from Pub/Sub to BigQuery. This is not only because both Pub/Sub and Dataflow are serverless tools that help reduce maintenance costs, but Dataflow also offers real-time streaming processing capabilities, such as watermark and programmable windowing, to ensure the freshness and correctness of the processed data.

"Pub/Sub and Dataflow together work as a powerful toolset for streaming data processing that remains serverless in the core. These conveniences save us time to focus on process logic development and boost productivity."

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To gauge the efficacy of the approach, GoGoX employs key metrics, such as assessing the extent of subsidies required for OA drivers who fall short of the guaranteed income threshold, tracking the overall number of GGB orders they successfully complete, and evaluating the financial outlay in GGB order incentives. These metrics not only provide insights into the financial well-being of GoGoX's drivers but also serve as crucial indicators of the overall success of the company's OA system. Its new OA algorithm is able to decrease driver subsidy by 42%, increase GGB orders served by OA drivers by 96%, and thus decrease total GGB incentive by 30%. In the future, by leveraging CDC pipeline, GoGoX will continuously train its ML models with the latest data in real time and realize predictive data analytics with to-the-minute accuracy.

"We wanted to make our CDC jobs economically viable and reduce the maintenance cost," explains Tsai. "Pub/Sub and Dataflow together work as a powerful toolset for streaming data processing that remains serverless in the core. These conveniences save us time to focus on process logic development and boost productivity."

Enhancing user experience with voice chatbots and more reliable cloud services

GoGoX's effort to enhance user experience continues. The team is currently developing voice chatbots for customer service and sales calls with the generative AI technology of Google, Speech-to-Text (https://cloud.google.com/speech-to-text), Text-to-Speech AI (https://cloud.google.com/text-to-speech), Dialogflow (https://cloud.google.com/dialogflow), and Vertex AI (https://cloud.google.com/vertex-ai). It is also considering adopting Anthos (https://cloud.google.com/anthos) to enhance disaster recovery across its public cloud services, and fortifying security with security products of Google Cloud.

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> logistics operations with real-time insights to ensure our customers' and drivers' needs are met. With the abundant tools provided by Google Cloud, we're well positioned to continue improving our services."
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